# ESE 3207 Concept Primer

Solar Photovoltaic Systems · KUET · read this first: the subject from zero, before the past-paper solutions

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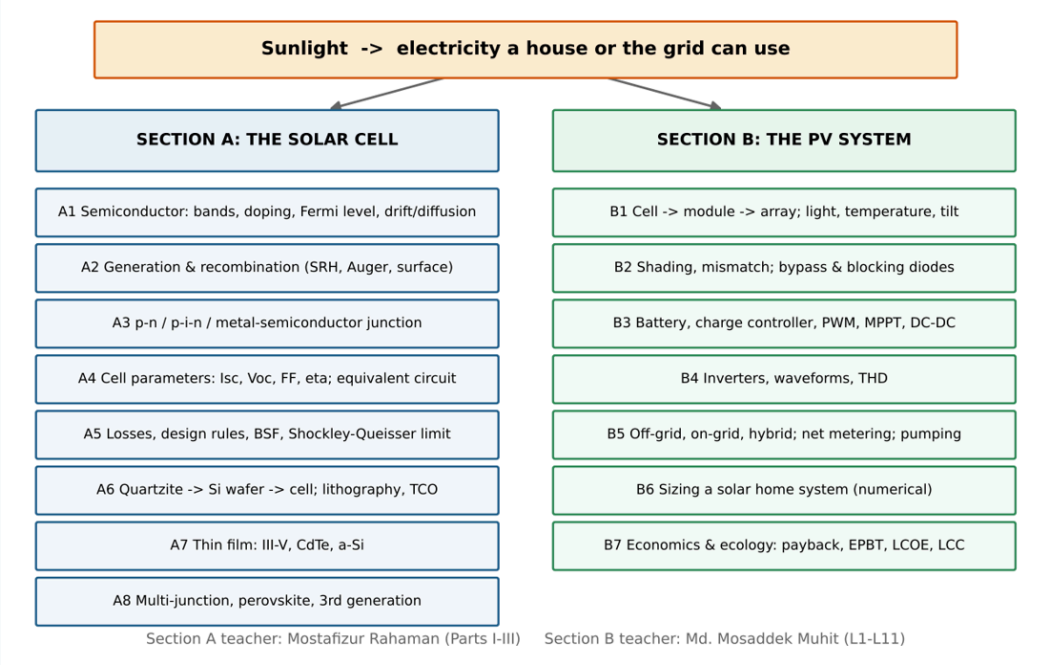

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## How to use this primer

This booklet teaches solar photovoltaics from zero, in the order the exam needs it. Read it once before the Complete Solutions. Each section has a green **core idea** box (learn this first), the key figure, the formulas with every symbol and unit, and a short worked example. Section A (cell physics) is mostly **explain with a diagram**, with a few short numericals (photocurrent, $V_{bi}$, W, carrier densities, fill factor). Section B (systems) is half descriptive and half numericals that repeat (temperature derating, inverter selection, SHS (solar home system) sizing, life-cycle cost).

**The pass set** (from the Question Map): A1-A6 and B1-B7; A7 (thin film) and A8 (multi-junction, perovskite, generations) are the next to add.

### The whole course on one page




Concept map of ESE 3207

Read the map from left to right. **Physics**: a semiconductor absorbs a photon and makes an electron-hole pair; a p-n junction separates the pair, giving current and voltage. **Device**: the cell is described by its I-V curve ($I_{sc}$, $V_{oc}$, FF (fill factor), η) and loses energy in known ways. **Technology**: silicon wafers, thin films and multi-junction cells. **System**: cells make modules and arrays; a controller, battery and inverter deliver the energy to loads or the grid; sizing and economics decide how big and whether it pays.

## Constants, units and maths refresher

> Keep these constants on your answer script; most Section A numericals use only them. Work in **cm** for semiconductor numbers (doping in $cm^{-3}$, ε in F/cm) and in **SI** for light ($W/m^2$, m).

| Quantity | Value |
|---|---|
| electron charge q | $1.602 \times 10^{-19}$ C (also 1 eV = $1.602 \times 10^{-19}$ J) |
| Planck h; speed of light c | $6.626 \times 10^{-34}$ J·s; $2.998 \times 10^{8}$ m/s; hc = 1240 eV·nm |
| thermal voltage kT/q at 300 K | 0.0258 V (k = $1.381 \times 10^{-23}$ J/K = $8.617 \times 10^{-5}$ eV/K) |
| permittivity $\varepsilon_0$; Si $\varepsilon_r$ | $8.854 \times 10^{-14}$ F/cm; 11.7 |
| Si at 300 K | $E_g$ = 1.12 eV, $n_i = 1.5 \times 10^{10}$ $cm^{-3}$ (some books $9.65 \times 10^{9}$), $N_C = 3.22 \times 10^{19}$, $N_V = 1.83 \times 10^{19}$ $cm^{-3}$ |
| STC (standard test conditions (1000 $W/m^2$, AM1.5, 25 °C)) | 1000 $W/m^2$, AM1.5 spectrum, 25 °C cell |
| length and area | 1 µm = $10^{-4}$ cm = $10^{-6}$ m; 1 nm = $10^{-9}$ m; 1 $cm^2$ = $10^{-4}$ $m^2$; 1 $mA/cm^2$ = 10 $A/m^2$ |

### Exponentials and logarithms

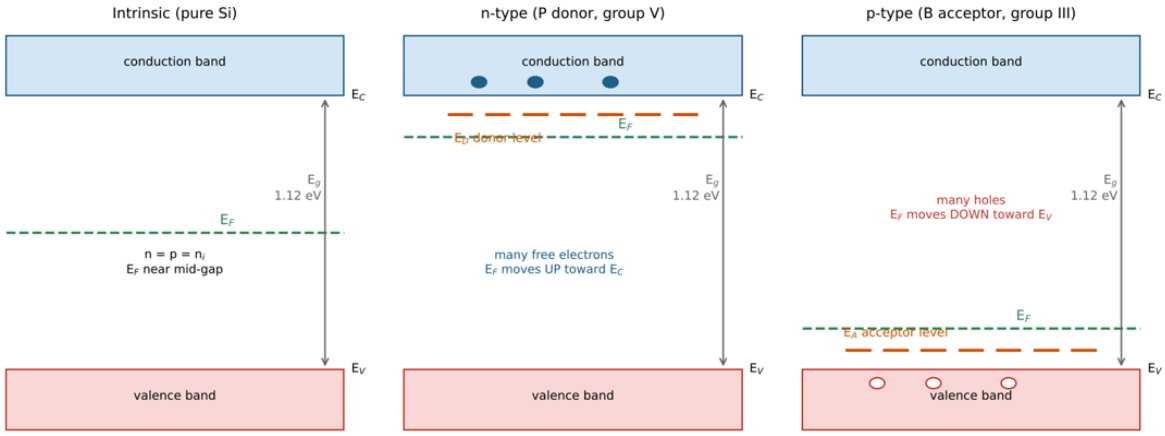
- $e^x$ grows very fast: the diode current doubles for every 18 mV of extra forward voltage at 300 K. $e^{-x}$ decays fast: light intensity $e^{-\alpha x}$ falls to 37% after one absorption length $1/\alpha$, to 2% after $4/\alpha$.
- ln undoes e: if $I/I_0 = e^{qV/kT}$ then $V = (kT/q)\ \ln(I/I_0)$. ln(10) = 2.303, so $V_{oc}$ rises by 0.0258 × 2.303 = 59.5 mV for a tenfold increase in light.
- Powers of ten: $(1.5 \times 10^{10})^2 = 2.25 \times 10^{20}$; $10^{16} \times 10^{18} = 10^{34}$; divide exponents by subtracting.

### Energy, power and battery units

- Power (W) = V × I. Energy (Wh) = power × hours. 1 kWh = 1000 Wh = 3.6 MJ.
- Battery capacity (Ah) × voltage (V) = stored energy (Wh): 200 Ah × 12 V = 2.4 kWh. Usable energy = capacity × DOD (depth of discharge).
- Peak sun hours: daily solar energy in $kWh/m^2$ written as hours of 1000 $W/m^2$. 5 $kWh/m^2$/day = 5 PSH (peak sun hours), so a 100 Wp module gives about 500 Wh/day before losses.

## Semiconductors and doping (A1)

> In a crystal, electron energies form a **valence band** (bonded electrons) and a **conduction band** (free electrons), separated by the **bandgap $E_g$**. Breaking a bond makes a free electron and a **hole**. Adding phosphorus (5 valence electrons) gives extra electrons: **n-type**; adding boron (3) gives extra holes: **p-type**. The **Fermi level** $E_F$ shows where electrons are: near $E_C$ in n-type, near $E_V$ in p-type.



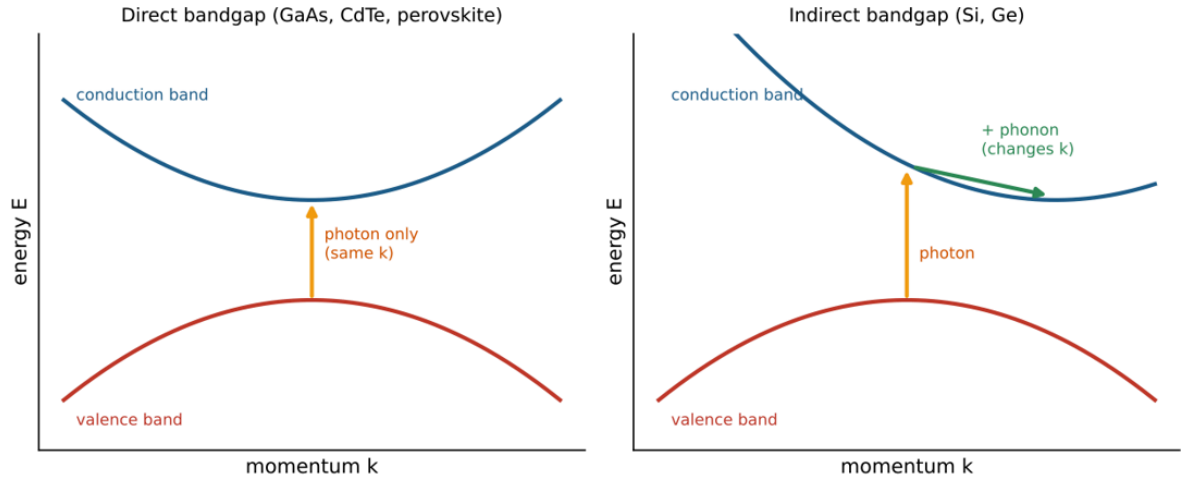

Intrinsic, n-type and p-type band diagrams

$$n \cdot p = n_i^2; \quad n = N_C e^{-(E_C - E_F)/kT}; \quad p = N_V e^{-(E_F - E_V)/kT}$$

| Symbol | Meaning | Unit |
|---|---|---|
| $n$, $p$ | electron and hole densities | $cm^{-3}$ |
| $n_i$ | intrinsic carrier density | $cm^{-3}$ |
| $N_D$, $N_A$ | donor and acceptor doping | $cm^{-3}$ |
| $N_C$, $N_V$ | effective density of states of CB and VB | $cm^{-3}$ |
| $E_C$, $E_V$, $E_F$ | conduction band edge, valence band edge, Fermi level | eV |

**Example:** $N_A = 10^{16}$: $p = 10^{16}$, $n = 2.25 \times 10^{20}/10^{16} = 2.25 \times 10^4$ $cm^{-3}$.

**Carriers move** by **drift** (pushed by an electric field, speed = μE) and **diffusion** (spreading from high to low concentration, D). They are linked: D = μkT/q.




Direct gap (GaAs) vs indirect gap (Si)

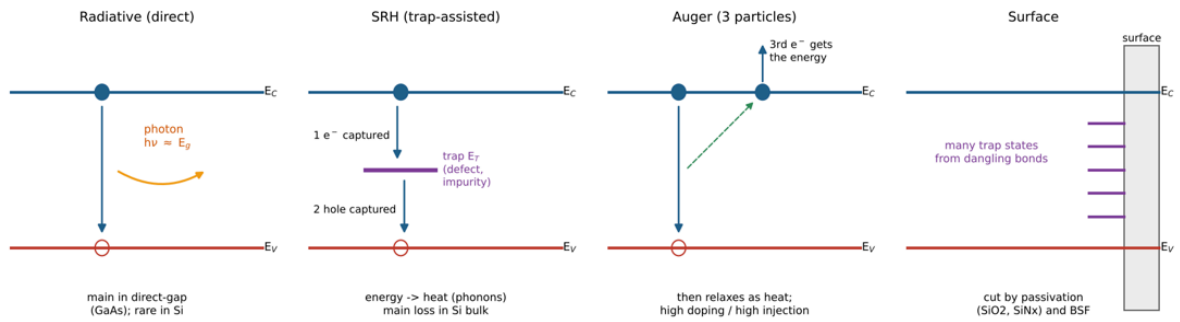
## Light in, carriers out: absorption and recombination (A2)

A photon with energy $E = hc/\lambda \geq E_g$ can be absorbed and make one electron-hole pair. Light intensity decays as $e^{-\alpha x}$. The extra carriers live for a **lifetime τ** and wander a **diffusion length $L = \sqrt{(D\tau)}$** before they recombine: radiatively (photon), via a defect (SRH, heat), by Auger (energy to a third carrier) or at the surface.

$E_{ph}$ (eV) = 1240/λ (nm); $\Phi_0 = P\lambda/(hc)$; $\Phi(x) = (1 - R)\Phi_0 e^{-\alpha x}$; $J_{ph}$ = q × (absorbed photon flux)

| Symbol | Meaning | Unit |
|---|---|---|
| λ | wavelength | nm or m |
| P | irradiance (light power per area) | $W/m^2$ |
| Φ | photon flux | $m^{-2}s^{-1}$ |
| R | front reflectance | fraction |
| α | absorption coefficient | $cm^{-1}$ or $m^{-1}$ |
| $J_{ph}$ | photocurrent density | $A/m^2$ |

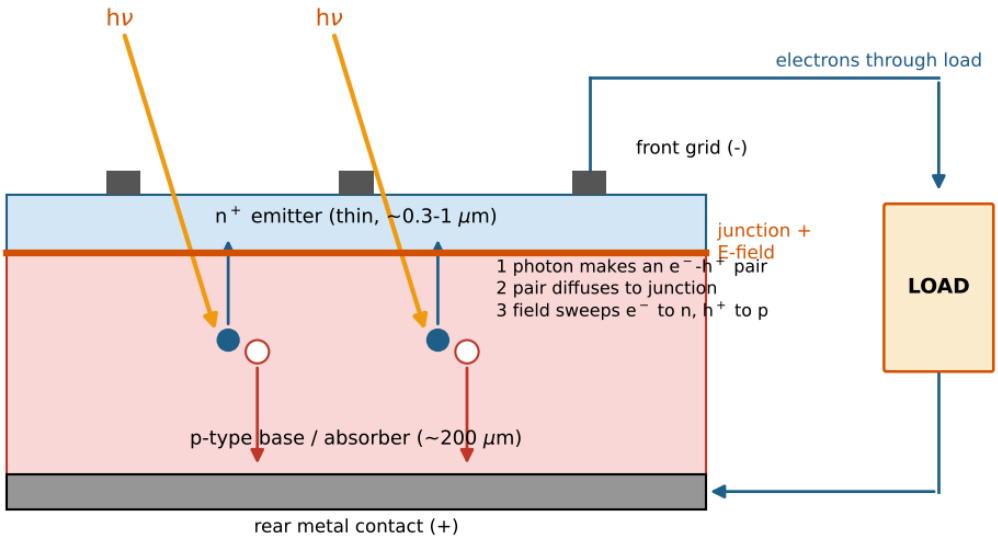
**Example:** 500 nm photon: 1240/500 = 2.48 eV (more than Si's 1.12 eV, so absorbed; the extra 1.36 eV becomes heat). 1000 $W/m^2$ of it is $2.52 \times 10^{21}$ photons/$m^2$s; with 38% reflected, $J = 1.602 \times 10^{-19} \times 0.62 \times 2.52 \times 10^{21} = 250$ $A/m^2$.




Recombination paths

## The p-n junction and the solar cell (A3)

Join p and n material: carriers diffuse across and leave behind fixed ions, forming a **depletion region** with a **built-in electric field** and voltage $V_{bi}$. Light-made electrons and holes that reach this field are separated: electrons go to n, holes to p. That separation is the photocurrent and the photovoltage.




How a silicon solar cell works

$V_{bi} = (kT/q)\ \ln(N_A N_D/n_i^2);\quad W = \sqrt{[(2\varepsilon/q)(V_{bi} - V)(1/N_A + 1/N_D)]};\quad E_{max} \approx 2V_{bi}/W$

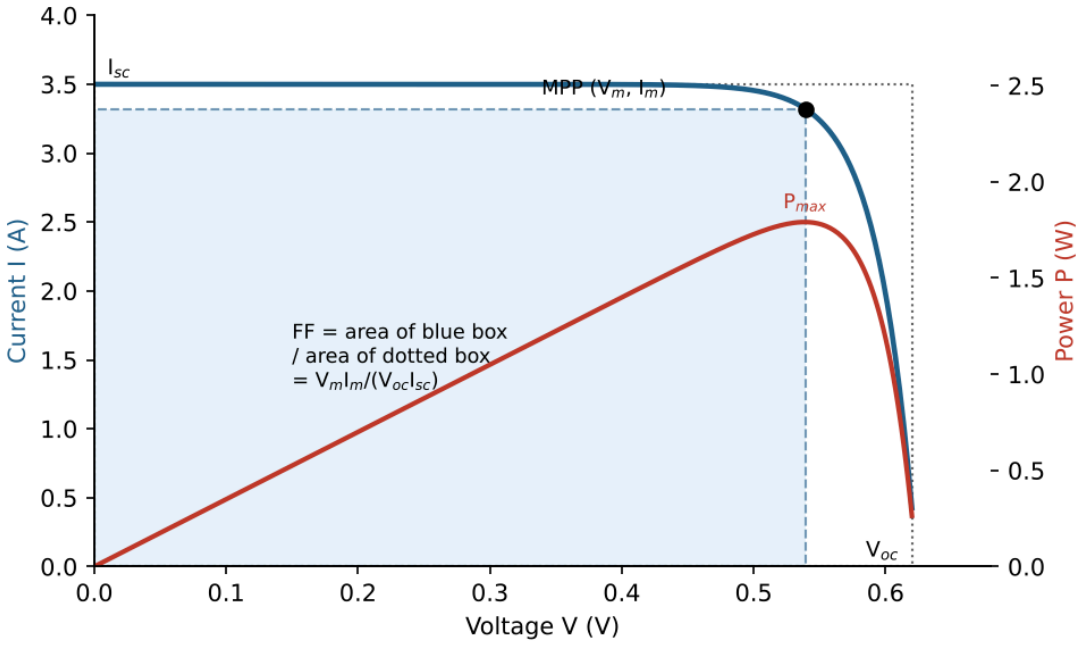
| Symbol | Meaning | Unit |
|---|---|---|
| $V_{bi}$ | built-in voltage | V |
| $W$ | depletion width | cm |
| $\varepsilon$ | $\varepsilon_r\varepsilon_0 = 1.036 \times 10^{-12}$ for Si | F/cm |
| $V$ | applied voltage (negative for reverse bias) | V |
| $E_{max}$ | peak electric field | V/cm |

**Example:** $N_A = 10^{16}$, $N_D = 10^{18}$: $V_{bi} = 0.0258\ \ln(10^{34}/2.25 \times 10^{20}) = 0.81$ V; $W = 0.325$ μm; $E_{max} = 5 \times 10^4$ V/cm.

In a **p-i-n** cell an undoped layer sits between p and n; the field fills it, so carriers are collected by drift (needed for amorphous Si and perovskites, whose diffusion lengths are short).

## The I-V curve and the equivalent circuit (A4)

A solar cell is a **current source** (light) in parallel with a **diode** (the junction). Short-circuit gives $I_{sc}$; open circuit gives $V_{oc}$; in between there is one point of maximum power. The **fill factor** says how square the curve is. Real cells add a series resistance (lowers FF near $V_{oc}$) and a shunt resistance (lowers FF near $I_{sc}$).

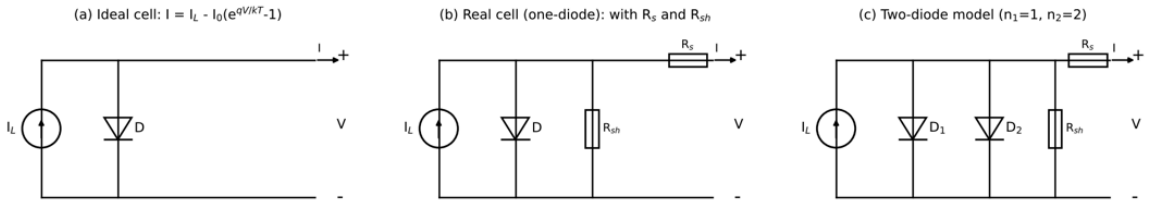



I-V and P-V curves

$$I = I_L - I_0[e^{q(V+IR_s)/nkT} - 1] - (V + IR_s)/R_{sh};\quad V_{oc} = (kT/q)\ln(I_L/I_0 + 1);\quad FF = V_m I_m/(V_{oc} I_{sc});\quad n = V_{oc} I_{sc} FF/(GA)$$

| Symbol | Meaning | Unit |
|---|---|---|
| $I_L$ | light-generated current ($\approx I_{sc}$) | A |
| $I_0$ | dark saturation current | A |
| n | ideality factor (1-2) | – |
| $R_s$, $R_{sh}$ | series and shunt resistance | Ω |
| $V_m$, $I_m$ | voltage and current at the maximum power point | V, A |
| G | irradiance | $W/m^2$ |
| A | cell or module area | $m^2$ |

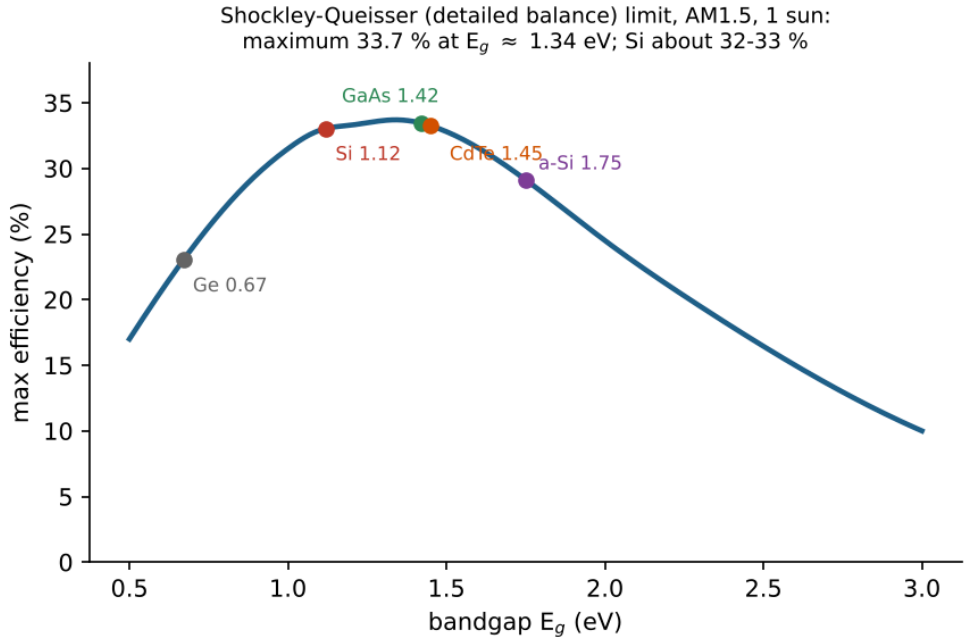
**Example:** $V_{oc}$ = 0.6 V, $I_{sc}$ = 3.5 A, $V_m$ = 0.5 V, $I_m$ = 3.1 A, 100 $cm^2$: $P_m$ = 1.55 W; FF = 1.55/2.1 = 0.738; η = 1.55/(1000 × 0.01) = 15.5%.




Equivalent circuits

## Losses and limits (A5)

Photons below $E_g$ pass through (**non-absorption**); photons above $E_g$ waste their extra energy as heat (**thermalisation**). These two set the **Shockley-Queisser limit**: about 33.7% at $E_g$ = 1.34 eV for one junction. Practical losses add on top: reflection, shading by fingers, recombination, resistance.

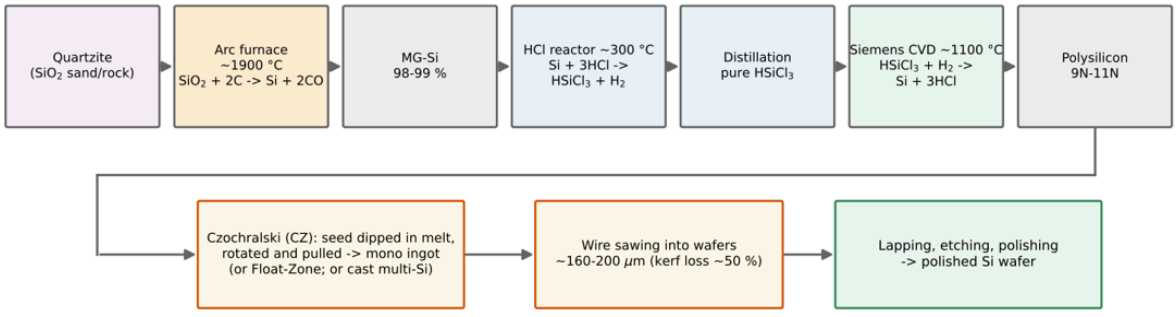



Shockley-Queisser limit

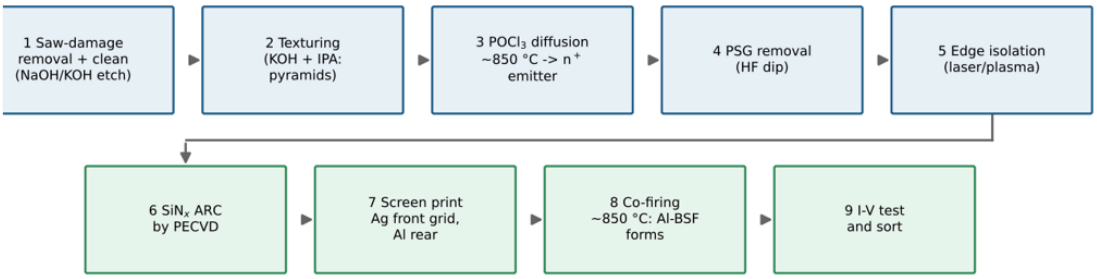
The cures map one-to-one: texture + ARC (reflection), thin fingers (shading), passivation + BSF (recombination), heavy doping under contacts (resistance), multi-junction (both spectral losses).

## Making silicon cells (A6)

Sand/quartz → arc furnace (MG-Si, 98%) → trichlorosilane → distillation → Siemens reactor (polysilicon, 9N) → Czochralski ingot → wire-sawn wafers → texture, phosphorus diffusion, $SiN_x$ ARC (anti-reflection coating), screen-printed contacts, firing → tested cell.



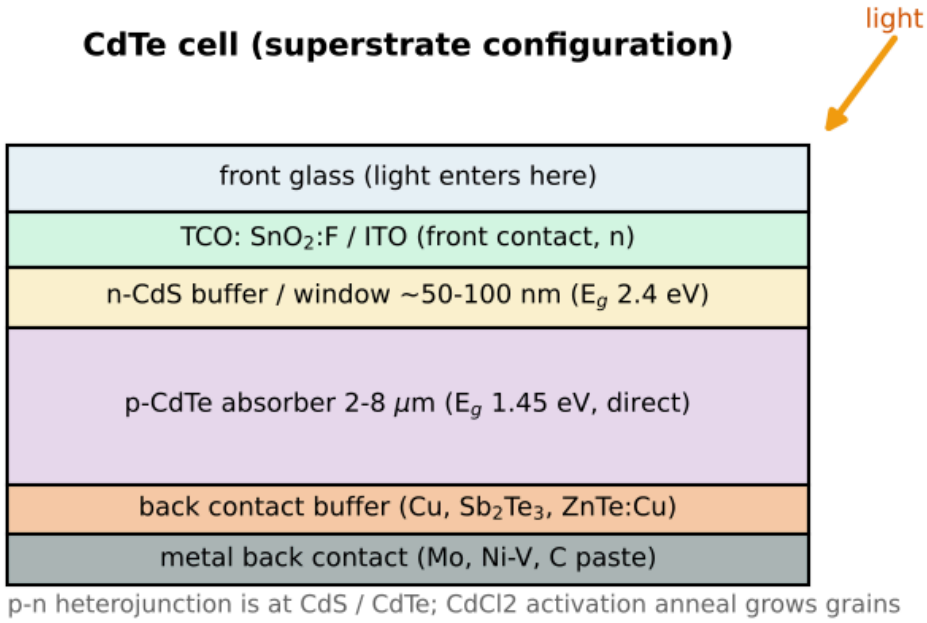

From quartzite to wafer



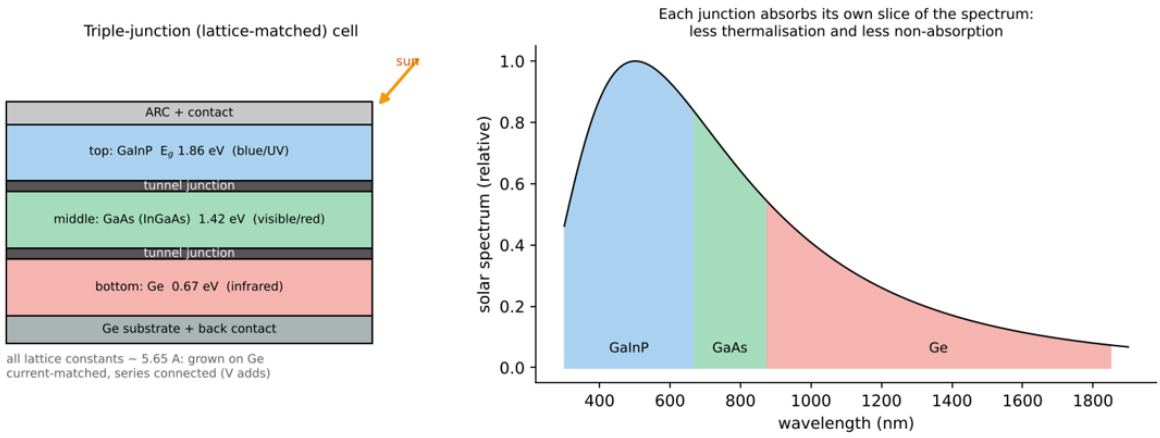

Cell process flow

## Thin films and advanced cells (A7, A8)

> **Direct-gap** materials (CdTe, CIGS (copper indium gallium diselenide), GaAs, perovskite) absorb sunlight in 1-3 µm, so cells can be thin films on glass. **Multi-junction** cells stack several gaps (largest on top) so each photon is used by a junction just below its energy; they must be matched in bandgap, current and lattice constant.



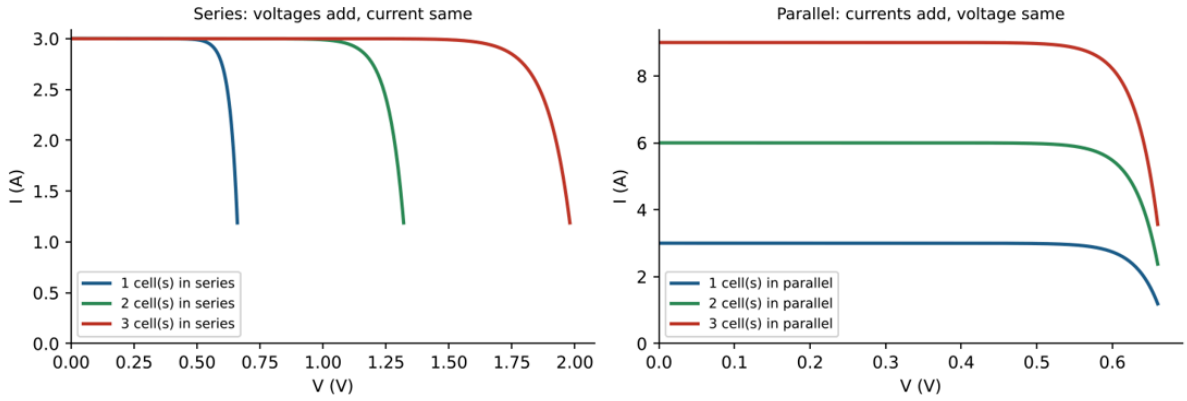

CdTe thin-film cell




Triple-junction cell

## From cell to array; light and heat (B1)

Series: voltages add, current stays. Parallel: currents add, voltage stays. Current is proportional to light and area; voltage falls as the cell heats up (about −2.3 mV/°C per Si cell), so power falls about 0.45% per °C above 25 °C.




Series and parallel I-V curves

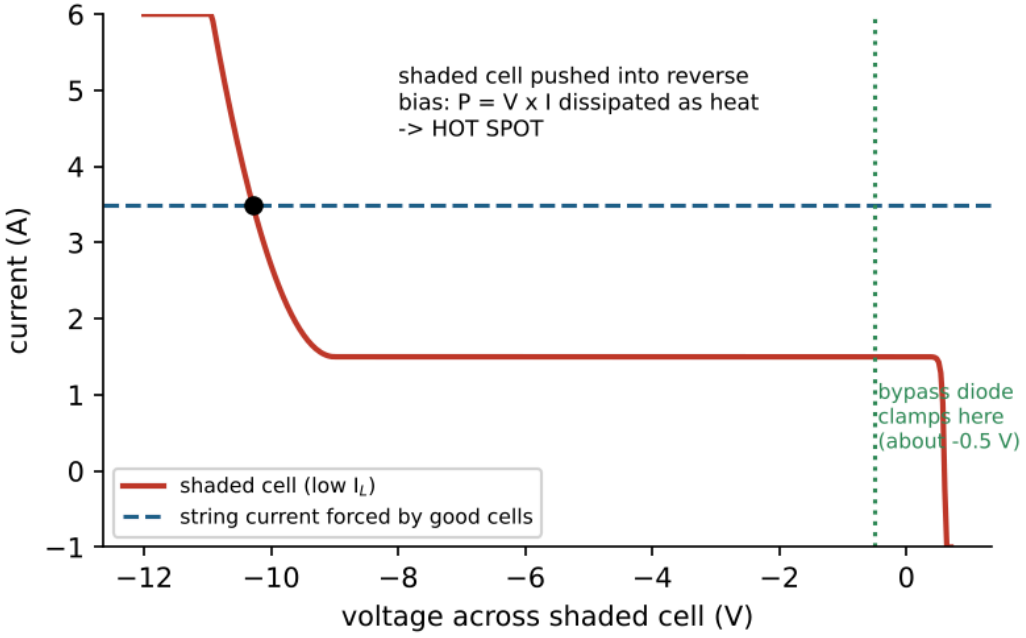
$P(T) = P_{STC}[1 + \gamma(T - 25)]$; $I_{sc}(G) = I_{sc,STC} \times G/1000$; $N_s = V_{needed}/V_{m,cell}$; $N_p = I_{needed}/I_m$

| Symbol | Meaning | Unit |
|---|---|---|
| $\gamma$ | temperature coefficient of power (−0.0045 /°C) | 1/°C |
| $T$ | cell temperature | °C |
| $N_s$, $N_p$ | number in series and in parallel | – |

**Example:** 75 Wp at 45 °C: 75 × (1 − 0.0045 × 20) = 68.25 W.

## Shading and diodes (B2)

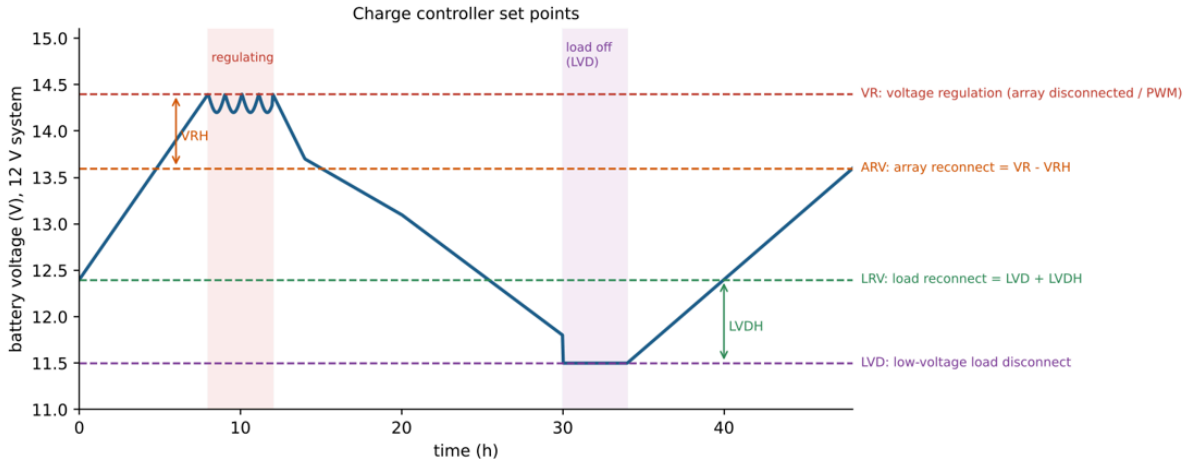
In a series string, the shaded cell sets the current for everyone, and it can be forced into reverse bias and become a **hot spot**. A **bypass diode** (in parallel, opposite polarity) gives the current a way around; a **blocking diode** (in series) stops the battery discharging into the panel at night.




Hot spot and bypass diode

## Battery, charge controller, MPPT (maximum power point tracking (tracker)) and converters (B3)

> The charge controller keeps the battery between a high limit (stop charging) and a low limit (cut the load). A PWM (pulse width modulation) controller pulses the current near full charge. An MPPT controller is a DC-DC converter that keeps the panel at its maximum power point by matching impedance.




Controller set points

`Buck: $V_{out} = DV_{in}$;  Boost: $V_{out} = V_{in}/(1 - D)$;  at MPP (maximum power point) dP/dV = 0 ⇔ dI/dV = -I/V;  C-rate current = Ah/h`

| Symbol | Meaning | Unit |
|---|---|---|
| D | duty cycle = on-time/period | – |
| Ah | battery capacity | A·h |
| DOD | depth of discharge: fraction of capacity used | – |

## Inverters (B4)

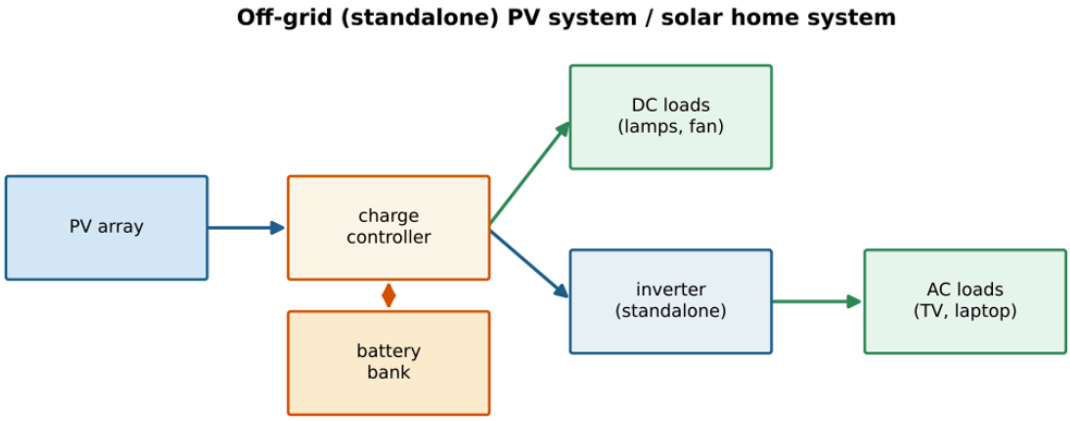
An inverter turns DC into AC with a bridge of switches. Square wave is cheap but distorted; modified sine is in between; pure sine (PWM + filter) is clean and required for the grid. THD (total harmonic distortion) measures the distortion.

$P_{out} = nP_{in}$; loss = $P_{in} - P_{out}$; $I_{DC} = P_{in}/V_{battery}$; rating ≈ 1.25 × load; $THD = \sqrt{(\Sigma V_h^2)}/V_1$

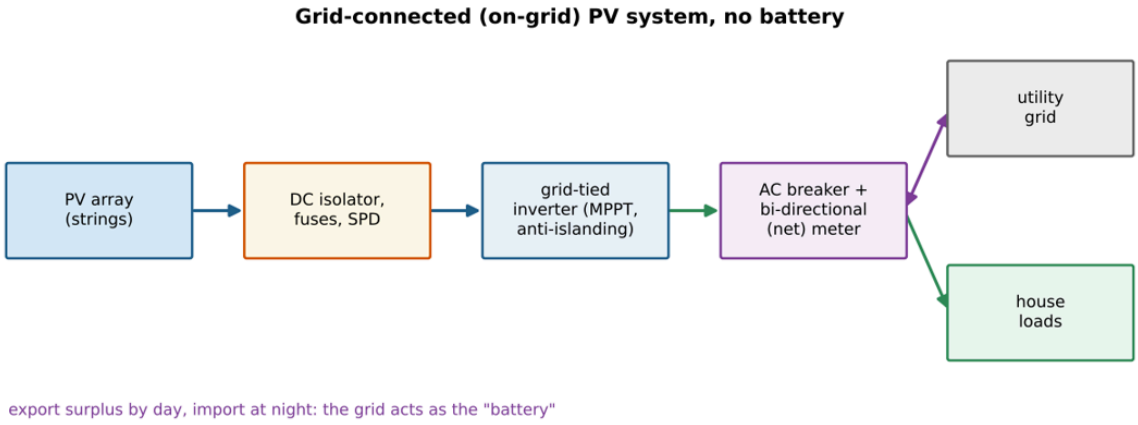
**Example:** 300 W in at 88%: 264 W out, 36 W lost.

## System types and net metering (B5)

**Off-grid**: PV, controller, battery, inverter, loads. **On-grid**: PV, grid-tied inverter, net meter, grid (the grid is the battery). **Hybrid**: add a battery or a diesel/wind generator. **Net metering**: the meter runs both ways and you pay only for import minus export.




Off-grid system




On-grid system

## Sizing a solar home system (B6)

> Size in this order: daily energy of the loads → PV watts → battery Ah → controller amps → inverter VA → switches → cables. Always add the 25% safety margin for the electronics and protection.

$E = \Sigma P \cdot n \cdot h$; $P_{PV} = E/(n \cdot PSH)$; $C_{bat} = E \cdot days/(V \cdot DOD)$; $I_{cc} = 1.25\ P_{PV}/V$; $S_{inv} = 1.25\ P_{load}$; $A = 2\rho LI/\Delta V$

| Symbol | Meaning | Unit |
|---|---|---|
| $E$ | daily load energy | Wh/day |
| $PSH$ | peak sun hours | h/day |
| $n$ | battery charging (system) efficiency | – |
| $C_{bat}$ | battery capacity | Ah |
| $\rho$ | copper resistivity 0.0175 | $\Omega \cdot mm^2/m$ |
| $L$ | one-way cable length | m |
| $\Delta V$ | allowed voltage drop (2-3% of system V) | V |
| $A$ | conductor cross-section | $mm^2$ |

**Example:** 1 kWh/day, 5 PSH, 80%: $P_{PV}$ = 1000/(0.8 × 5) = 250 Wp; battery for 2 days at 12 V, 50% DOD: 1000 × 2/(12 × 0.5) = 333 Ah.

## Economics and ecology (B7)

> Money spent in future years is worth less today. Convert every future cost to **present worth**, add them up (life-cycle cost), and compare systems. **LCOE** (levelized cost of electricity) is that cost per kWh. **Energy payback time** is how long the panel takes to make back the energy used to produce it (1-3 years).

$x = (1 + i)/(1 + d)$; $P_r = x^n$; $P_a = (1 - x^n)/(1 - x)$; $ALCC$ (annualised life-cycle cost) = $LCC$ (life-cycle cost)$/P_a$; $EPBT$ (energy payback time) = $E_{invested}/E_{per\ year}$

| Symbol | Meaning | Unit |
|---|---|---|
| $i$ | inflation rate | fraction/yr |
| $d$ | discount rate | fraction/yr |
| $n$ | year of the cost, or lifetime | yr |
| $P_r$ | present-worth factor of a one-time cost in year n | – |
| $P_a$ | present-worth factor of a yearly cost for n years | – |

**Example:** i = 3%, d = 10%: x = 0.9364; a battery costing $900 in year 5 is worth $900 \times 0.9364^5$ = $648 today.

# Abbreviations used in this pack

*Every short form, written out. The answers also spell each one out the first time it appears.*

| Short form | Full meaning |
|---|---|
| **a-Si:H** | hydrogenated amorphous silicon |
| **Ah** | ampere-hour |
| **ALCC** | annualised life-cycle cost |
| **AM** | air mass |
| **ARC** | anti-reflection coating |
| **ARV** | array reconnect voltage |
| **AZO** | aluminium-doped zinc oxide |
| **BOS** | balance of system |
| **BSF** | back surface field |
| **CB** | conduction band |
| **CIGS** | copper indium gallium diselenide |
| **CZ** | Czochralski |
| **CZTS** | copper zinc tin sulfide |
| **DOD** | depth of discharge |
| **EPBT** | energy payback time |
| **EQE** | external quantum efficiency |
| **ETL** | electron transport layer |
| **EVA** | ethylene vinyl acetate (encapsulant) |
| **FA** | formamidinium |
| **FF** | fill factor |
| **FTO** | fluorine-doped tin oxide |
| **FZ** | float zone |
| **HIT** | heterojunction with intrinsic thin layer |
| **HTL** | hole transport layer |
| **HVD** | high-voltage disconnect |
| **IBC** | interdigitated back contact |
| **IEEE** | Institute of Electrical and Electronics Engineers |
| **IncCond** | incremental conductance |
| **IQE** | internal quantum efficiency |
| **ITO** | indium tin oxide |
| **LCA** | life-cycle assessment |
| **LCC** | life-cycle cost |
| **LCOE** | levelized cost of electricity |
| **LRV** | load reconnect voltage |
| **LVD** | low-voltage disconnect |
| **LVDH** | low-voltage disconnect hysteresis |

| | |
|---|---|
| **MA** | methylammonium |
| **MBE** | molecular beam epitaxy |
| **MEG** | multiple exciton generation |
| **MG-Si** | metallurgical-grade silicon |
| **MOCVD** | metal-organic chemical vapour deposition |
| **MPP** | maximum power point |
| **MPPT** | maximum power point tracking (tracker) |
| **nc-Si:H** | hydrogenated nanocrystalline silicon |
| **NOCT** | nominal operating cell temperature |
| **P&O** | perturb and observe |
| **PBT** | payback time |
| **PECVD** | plasma-enhanced chemical vapour deposition |
| **PERC** | passivated emitter and rear cell |
| **PERL** | passivated emitter, rear locally diffused |
| **PID** | potential-induced degradation |
| **PSG** | phosphosilicate glass |
| **PSH** | peak sun hours |
| **PV** | photovoltaic |
| **PV/T** | photovoltaic-thermal (hybrid collector) |
| **PWM** | pulse width modulation |
| **SHS** | solar home system |
| **SOC** | state of charge |
| **SPV** | solar photovoltaic |
| **SPWM** | sinusoidal pulse width modulation |
| **SQ** | Shockley-Queisser |
| **SREDA** | Sustainable and Renewable Energy Development Authority (Bangladesh) |
| **SRH** | Shockley-Read-Hall (trap-assisted recombination) |
| **STC** | standard test conditions (1000 $W/m^2$, AM1.5, 25 °C) |
| **TCO** | transparent conducting oxide |
| **TCS** | trichlorosilane ($HSiCl_3$) |
| **THD** | total harmonic distortion |
| **TOPCon** | tunnel oxide passivated contact |
| **VA** | volt-ampere |
| **VB** | valence band |
| **VFD** | variable frequency drive |
| **VR** | voltage regulation (set point) |

| | |
|---|---|
| **VRH** | voltage regulation hysteresis |
| **Wp** | watt-peak (power at STC) |